THE FLAT AT A POINT FUNCTOR IS UNCHANGED BY A FINITE LOCAL EXTENSION CARRYING AN ISOMORPHIC MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The situation defining the flat at a point functor. Same chapter flat.tex, situation labelled situation-flat-at-point (statement, proof lines 6069 and 6079). Fixes the category over which the functors are defined and the notation for the base changes of the algebras and modules.
- Flatness of a module over a local ring is detected at maximal ideals of the localisation. Algebra, label algebra-lemma-flat-localization, invoked at proof line 6097. Allows the flatness of the localised module to be checked after localising further at maximal ideals, which is where the finite extension lets the maximal ideals be compared.
- The maximal ideals of a finite local extension correspond to the primes over the given prime. Algebra, label algebra-lemma-integral-going-up, invoked at proof line 6101. Identifies the maximal ideals of the localised extension with the primes lying over the given prime, so that the vanishing locus computed upstairs matches the image of the one downstairs.

Source reference. Original lemma 00HT: sources/stacks-chow/tags.txt, source lines 655–655. Navigation pointer for \ref{algebra-lemma-flat-localization} used inside the counted span (line 6097); the argument that consumes it is retained in the span.

Source reference. Original lemma 00GU: sources/stacks-chow/tags.txt, source lines 620–620. Navigation pointer for \ref{algebra-lemma-integral-going-up} used inside the counted span (line 6101); the argument that consumes it is retained in the span.

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Source reference. Original reference 05MK: sources/stacks-flat/pinned-original-source.tex, source lines 6008–6008. Navigation pointer for \ref{situation-flat-at-point} used inside the counted span (line 6069); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MK. Let $B \to C$ is a local map of local A-algebras and N a C-module. Denote $F'_{lf} : \mathcal{C} \to Sets$ the functor associated to the pair (C, N). If $M \cong N$ as B-modules and $B \to C$ is finite, then $F_{lf} = F'_{lf}$.*

Proof. Let A' be an object of $\mathcal{C}$. Set $C' = C \otimes_A A'$ and $N' = N \otimes_A A'$ similarly to the definitions of B', M' in Situation 05MK. Note that $M' \cong N'$ as B'-modules. The assumption that $B \to C$ is finite has two consequences: (a) $\mathfrak{m}_C = \sqrt{\mathfrak{m}_B C}$ and (b) $B' \to C'$ is finite. Consequence (a) implies that

$$V(\mathfrak{m}_{A'} C' + \mathfrak{m}_C C') = (\mathrm{Spec}(C') \to \mathrm{Spec}(B'))^{-1} V(\mathfrak{m}_{A'} B' + \mathfrak{m}_B B').$$

Suppose $\mathfrak{q} \subset V(\mathfrak{m}_{A'} B' + \mathfrak{m}_B B')$. Then $M'_\mathfrak{q}$ is flat over A' if and only if the $C'_\mathfrak{q}$-module $N'_\mathfrak{q}$ is flat over A' (because these are isomorphic as A'-modules) if and only if for every maximal ideal $\mathfrak{r}$ of $C'_\mathfrak{q}$ the module $N'_\mathfrak{r}$ is flat over A' (see Algebra, Lemma 00HT). As $B'_\mathfrak{q} \to C'_\mathfrak{q}$ is finite by (b), the maximal ideals of $C'_\mathfrak{q}$ correspond exactly to the primes of C' lying over $\mathfrak{q}$ (see Algebra, Lemma 00GU) and these primes are all contained in $V(\mathfrak{m}_{A'} C' + \mathfrak{m}_C C')$ by the displayed equation above. Thus the result of the lemma holds. □